



SESSION 3: COMPETING IN CONCENTRATED MARKETS MICROSOFT VS. GOOGLE

Competitive Strategy

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OUTLINE

- Concentrated markets and game theory
 - What is game theory?
 - How will we use it?
- Rivalry game -- The problem: Too much competition
 - Even though it destroys profits!!
- The solution: repetition

OLIGOPOLY AND GAME THEORY

- Extreme market structures:
 - Monopoly: no rivals
 - Perfect competition => No need to consider rivals' reactions
- Intermediate case:
 - Oligopoly (Greek=few sellers) or concentrated industry=> Need to consider rivals' reactions
- Game Theory:
 - formal study of strategic behavior, relation between inter-dependent agents (firms, trade authorities, armies, litigation...)

GAME THEORY

- Began in 1944 by von Neumann and Morgenstern.
 - Key development: in conventional economics market structure given.
 - People and firms are like machines, optimize given prices.
- With few players, need to consider interdependencies.
 - Value created by combinations of players

WHAT IS GAME THEORY?

- Anticipate rival's action
 - How will they choose
 - How will they react to my decisions
- Game theory: formal analysis of strategic interactions
- Applicability:
 - not set up model and solve;
 - rather, get insight
 - into the strategic interaction
 - into effect of changing the game

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RIVALRY GAME EXAMPLE: FASHION PRICING

- Use real stores, prices and quantities, and consider asymmetric payoffs
 - You are managing a line of fashion garment.
 - End of the season
 - Must discount to liquidate stock.
 - Decision: start discounting now or wait.
 - Only one competitor who is in the same situation as you.
 - fashion sensitive customers buy only the items they like
 - Estimate: 2m additional profits from fashion customers if you keep prices high only 1m if you discount now.
 - bargain hunters buy any discounted item.
 - 4m profits from the bargain hunters if you discount now and your competitor waits,
 - 2m if you and your competitor discount at the same time,
 - nothing from them if you wait and your competitor discounts now.

THE FASHION PRICING GAME

FORMAL APPROACH

		Store 2	
		Wait	Discount
Store 1	Wait	4	5
	Discount	5	3

Arrows indicate best responses for each player:

- From (Wait, Wait) to (Wait, Discount)
- From (Wait, Discount) to (Discount, Discount)
- From (Discount, Wait) to (Discount, Discount)

(Numbers in the table are profits)

GAME: A

MODEL OF STRATEGIC INTERACTIONS

What is a game?

- *Players* (e.g. Store 1 and Store 2)
- *Rules* (e.g. simultaneously choose whether to start discounting)
- *Strategies* (Discount or wait)
- *Payoffs* (Profits from strategy)

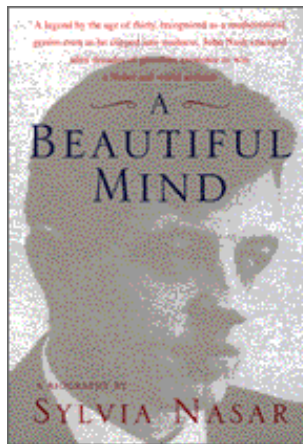
		Store 2	
		Wait	Discount
Store 1	Wait	4 4	2 5
	Discount	5 2	3 3

EQUILIBRIUM

CONCEPT OF REASONABLE OUTCOME

Nash Equilibrium= concept of
“reasonable” outcome

<http://www.nobel.se/economics/laureates/1994/index.html>



Store 1

Store 2

Wait ——— Discount

Wait

Discount

	Wait	Discount
Wait	4, 4	2, 5
Discount	5, 2	3, 3

Definition: Given what other players are doing, no player would want to change strategy *unilaterally* (i.e. each player's strategy is an optimal response to the other players' strategies)

TWO KEY CONCEPTS IN GAME THEORY

○ Nash Equilibrium

- Each player is doing best than he/she can, *given* the strategies of other players
 - Does not maximize joint profits!
 - Requires that reactions be mutually consistent

○ Dominant Strategy

- Player's strategy is best no matter what the others choose

SO WHAT IS THE PRICING PROBLEM?

Players have a dominant strategy – they should reduce prices

if you do not reduce– I am better off undercutting you and stealing market share

if you do – I am better off matching you

Regardless, we are likely to end up fighting even though it is in nobody's interest.

ANOTHER EXAMPLE: AIRBUS AND BOEING

	Price (m\$)	Orders	Order Share	Unit Cost (m\$)	Profits (b\$)
Low/Low	110	1000	500/500	100	5
High/High	115	800	400/400	100	6
Low/High	110/115	900	700/200	100	7/3

Airbus Price

Low High

	Low	High
Low	5	3
High	7	6

Boeing
Price

Other Examples

Choice of quantities by OPEC
members

Price setting at neighboring
petrol stations

Fare wars in airline industry

PRICING LIKE A PRISONER'S DILEMMA (PLAYERS A AND B)

	(b) confess	(b) deny
(a) confess	-10 -10	-25 -1
(a) deny	-1 -25	-3 -3

PRICE COMPETITION (BERTRAND GAME— WE DID IN FIRST CLASS)

- Firms compete on prices
- Assume all goods and consumers identical-- then undercutting by a little bit rival's price means stealing all the demand
- Then only Nash Equilibrium is $MC=P$ even with only two rivals
- If goods differentiated, cannot steal all demand, but similar game logic

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SOLUTION: COORDINATION AND DYNAMIC PRICING

- The point then is to manage to coordinate intention for low quantity
- If there is a long horizon of interactions in front, then it is possible to coordinate
 - In our game, last round completely uncooperative, even round before last
- Why?: key role of expectation of punishment

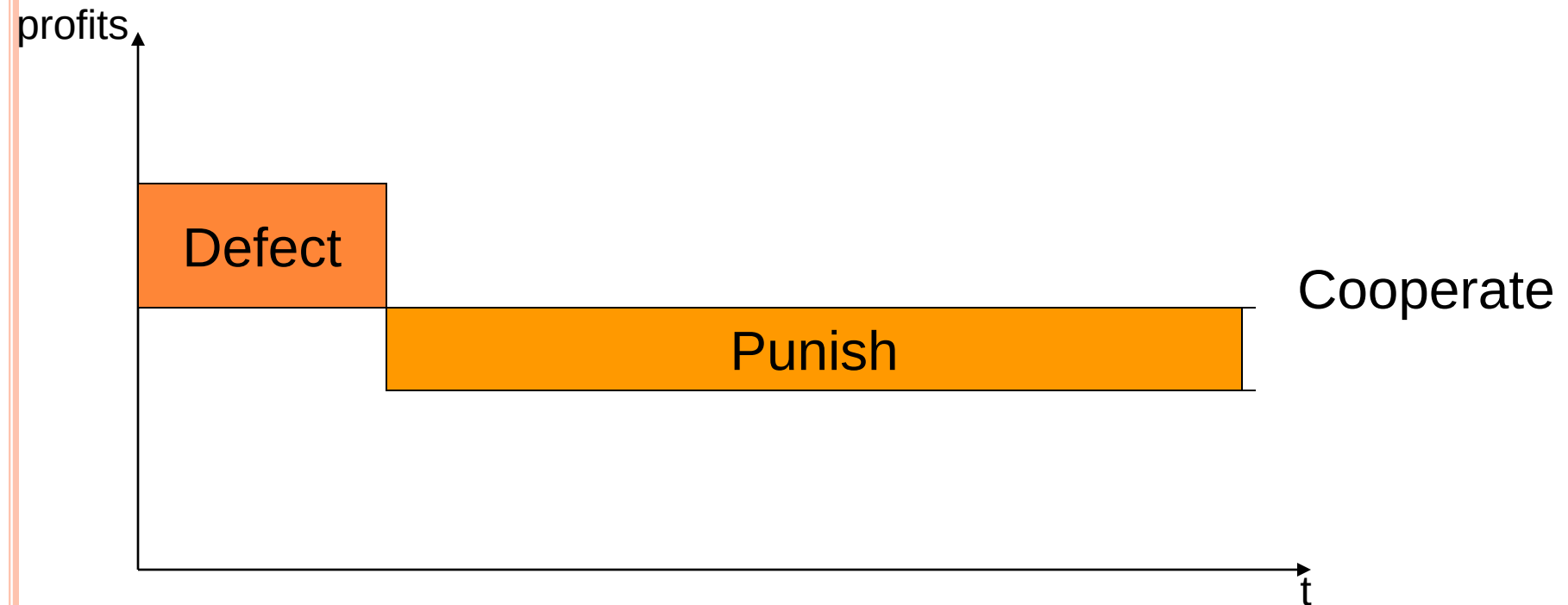
PUNISHMENT AND COORDINATION

- With repetition, coordination possible
 - If don't cheat, can charge monopoly price for ever, and get half of monopoly profit
 - If cheat, engage in price war, and get in low profit equilibrium for ever
- Note crucial role here of long horizon
 - Example: airlines that are on the verge of bankruptcy cut prices. Why? Future not valuable to them (they won't be around)

REPEATED PLAY CAN SUSTAIN COOPERATION

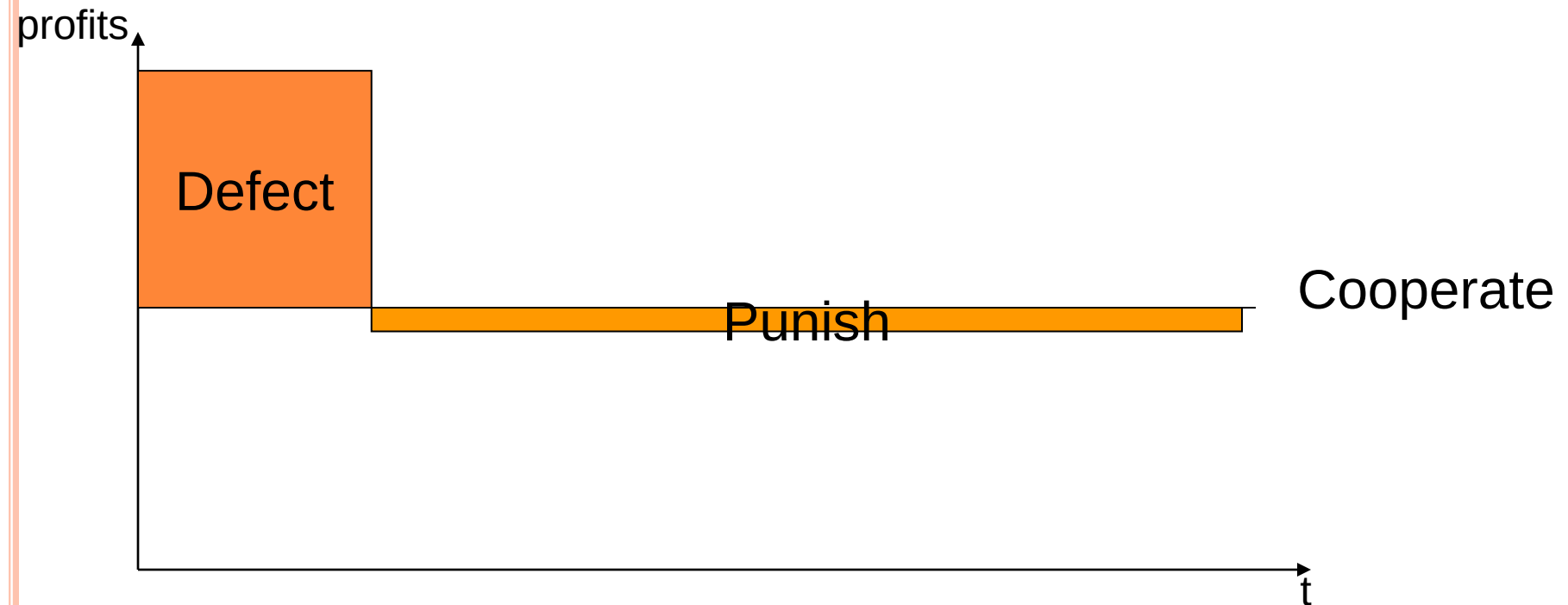
- As long as the gain from deviation and not being able to cooperate in the future (the loss if punish and cannot cooperate) is smaller than the gain from cooperation
- Can facilitate cooperation by making the deviation
 - Yield a small gain
 - Easy to detect
 - Incur in a heavy but credible punishment
 - Small relative to the future (e.g. value future a lot, there is growth so that future is worth a lot relative to the present, things are quite certain)

REPEATED PLAY



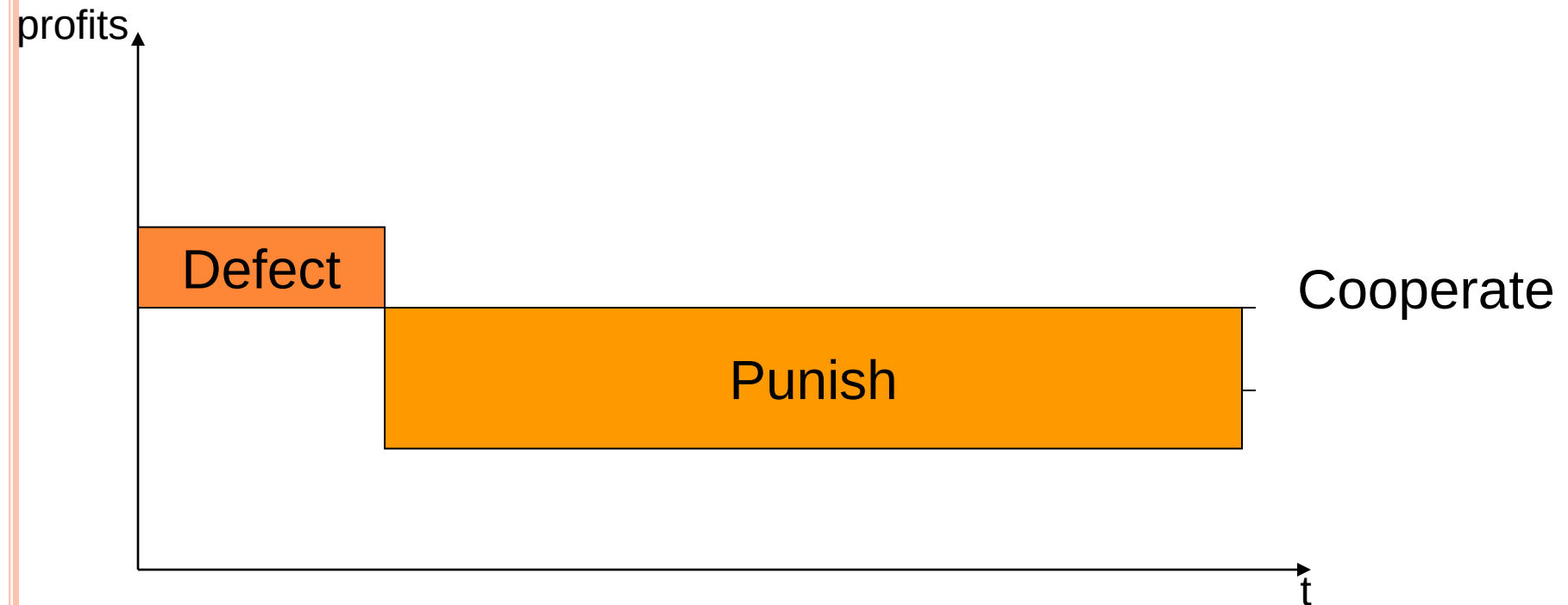
=> Worry about future cooperation keeps you 'honest'

REPEATED PLAY: DEVIATION ATTRACTIVE



⇒ Here you do not really have much to lose in terms of future cooperation so more likely to “cheat”

REPEATED PLAY: DEVIATING UNATTRACTIVE



=> Worry about future cooperation keeps you 'honest'

HOW DO YOU PROMOTE IMPLICIT COORDINATION?

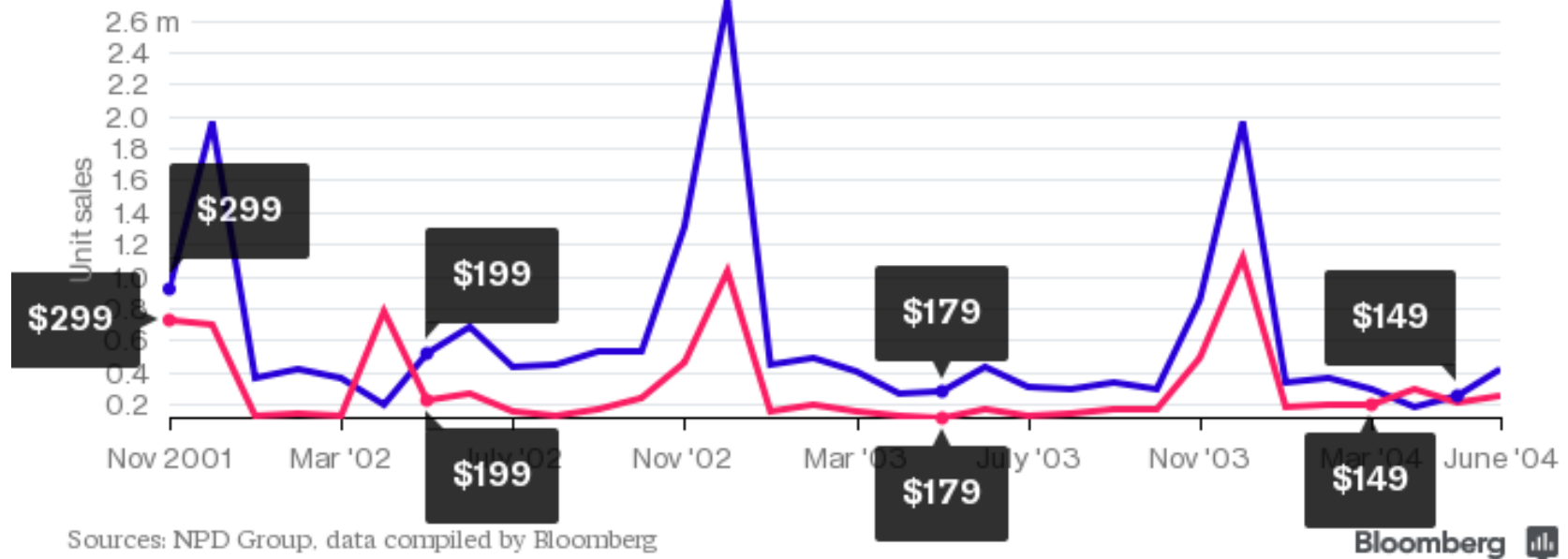
- Axelrod: tit-for-tat is (experimentally) best strategy to induce cooperation
 - Do what your rival did to you previously
- Examples:
 - Clause that 'we will not be undersold'
 - Airline industry in 1998 (see next slide)

- March Northwest begins sale on 14-day advance purchase fares. Others match
 - April Continental tries to raise leisure fares \$20 round-trip twice. Other carriers match, but Northwest blocks both attempts
 - May Northwest initiates 25 % sale on 21 -day advance-purchase fares. Others match. Later, Continental tries 5% increase on non-sale leisure fares. United, American, Delta and USAir all match, but not Northwest. Increase is rolled back
 - July: Continental raises leisure fares \$20 round-trip; Northwest blocks
- August 11 Delta and American raise fares by 4%, Northwest blocks
- August 14 American raises fares by 4% in markets not served by Northwest; carriers other than Northwest match
- August 17 Delta rolls back the increases; other airlines follow
- August 18 Northwest raises fares by 4%. All other carriers match the increase
- August 19 Northwest rescinds fare hike; US Airways and TWA match same day
- August 20 American, Delta, Continental match reversal; other carriers follow
- August 21 Northwest hikes fares by 4%, other carriers match

CONSOLE WARS

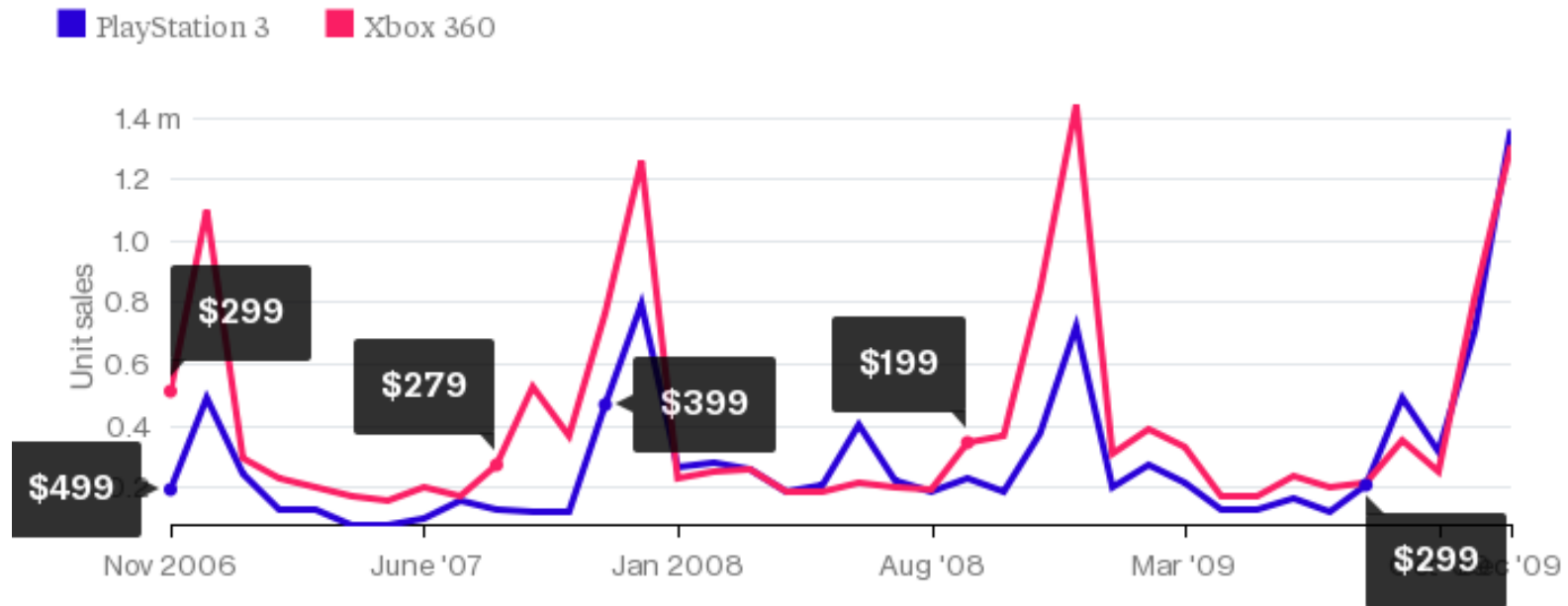
Tit for Tat

■ PlayStation 2 ■ Xbox



CONSOLE WARS 2:

Price Wars



Sources: NPD Group, data compiled by Bloomberg

Bloomberg

CONSOLE WARS (3): NO DIFFERENTIATION (SALES)

Game System Showdown



Sources: VGChartz, data compiled by Bloomberg

Bloomberg

TIT-FOR-TAT

- Why does it work? Axelrod argues:
 - Punishes firm that does not cooperate but does not punish for ever
 - Easy for rival to understand
 - Never defects first, in order not to provoke retaliation
- Problem: large cost of slight misperception
 - Any mistake 'echoes' back and forth
- A solution (DN:113): forgiving tit-for-tat
 - Wait until it is clear that defection taking place

INDUSTRY STRUCTURE CONDITIONS THAT FACILITATE IMPLICIT COOPERATION

- A small number of firms in market
- Large firms are unequally sized
- Public price or other sale conditions
- Firms are otherwise (culturally, technologically) similar
- Asset Specificity is low
- High capacity utilization
- Demand and costs are stable

FACILITATING PRACTICES

- Firms can undertake practices that facilitate implicit cooperation
- See book
 - Price leadership
 - Advance announcement of price changes
 - Most favored customer clauses
 - Meet the competition clause
 - Strategic use of inventories and backlogs
 - Buffering to avoid gaining market share when demand goes down

FIRM ASYMMETRIES AND PRICE UMBRELLAS

- If small firm undercuts large firm, large firm may decide not to respond (Airlines)
 - Matching is very costly (large number of units sold times the decrease in the margin)
 - Gain from matching is small (since rival is small)

FOCAL POINTS

- Problem with tacit cooperation: many long run equilibrium points. How to choose one in particular?
- Use a focal point: a point that industry consider reasonable
 - Example: Department stores mark-up of 200%.
 - OPEC: maintain certain historical market shares
- Criteria for choosing focal points: history, symmetry, efficiency...

Very important modern day analysis;

Mutually assured destruction.
first strike capability.
death hand.

EXAMPLE: AIRLINE INDUSTRY

- Airlines: industry analysis:
 - Demand volatile
 - Very high fixed cost, low marginal cost
 - Low BTE
 - No differentiation
 - Very hard to coordinate: multiple fares/non-transparent pricing
- Result: brutal price competition

How to know if party B is attacking me? when B reduce price.

CHANGING THE GAME: AMERICAN AIRLINES VALUE PRICING

- American Airlines introduced value pricing in April 1992 which simplifies tariffs
 - All fares built by distance x 4 fares
- This gives up price discrimination possibility- what is the point?

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- American airlines introduced value pricing in April 1992 which simplifies tariffs
 - All fares built by distance x 4 fares
- This gives up price discrimination possibility- what is the point?
 - Under old system, 93% of fares discounted
 - In effect, raises fare
 - But do not need to abandon price discrimination?

WHY DO IT?

- Simplify
- Facilitate price coordination
 - If all follow everyone better off

BUT

Northwest Airlines announcement (May 1992)

“KIDS FLY FREE”

What is NW hoping? its hoping that competitors do not copy.

AA

RESPONSE?

Tit for tat?

- Kinds fly free too?

NO!: GO NUCLEAR

50 % price cut on every seat every fare every flight

WHY NWA DID THIS?

- Target elastic segment
 - Great if rivals do not respond
- Asymmetries mean this works even if rivals respond
 - NWA was worst airline/lowest quality
 - If rivals are capacity constrained, NWA gains disproportionately from lower prices
- In general, worse players gain from extra demand coming in at lower prices
- Price wars tend to get started by smaller/worse players

TAKE AWAYS

- Concentrated markets and game theory
 - Oligopolistic markets different from perfect competition
 - Need to predict what your rival will do
 - Game theory is a tool to do this
- Rivalry game –
 - Like a prisoner's dilemma
 - The problem: more competition than either of players would like
- The solution: repetition
 - Dynamic strategies that are clear, imitable, and reduce incentive to deviate while increasing punishment